

4 Statistical physics – solutions

The problems are roughly in order of difficulty. Those indicated with ♡ are those which it is most important to work on. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

Problem 4.1 East, west, north, south

The number of atoms in one of the sub-volumes is binomially distributed with $p = 0.25$ and $N = 16$.

- (i) The mean number of atoms in a sub-volume is $\bar{n} = pN = 4$.
- (ii) The standard deviation is $\sigma = \sqrt{p(1-p)N} = \sqrt{0.25 \times 0.75 \times 16} = 1.732$.
- (iii) Now $p = 0.75$ and $q = 0.25$, so again $\sigma = 1.732$.

Problem 4.2 Large and small partitions

This is another binomial distribution, with $p = \frac{V}{V_0}$ and $q = (1 - \frac{V}{V_0})$. The total number of atoms is N , therefore the mean number is $\bar{n} = pN = N\frac{V}{V_0}$ and the standard deviation is $\sigma = \sqrt{pqN}$.

- (i) For $V \ll V_0$, $q \sim 1$ and so $\bar{n} = N\frac{V}{V_0}$ and $\sigma = \sqrt{pqN} = \sqrt{N\frac{V}{V_0}}$.
- (ii) For $V \rightarrow V_0$, $\bar{n} \rightarrow N$, $p \rightarrow 1$ and $q \rightarrow 0$ so $\sigma \rightarrow 0$.

Problem 4.3 Counting partitions

There are 7 ways (available microstates). Taking the atoms in turn, we can have:

1ε	1ε	1ε
2ε	1ε	0
2ε	0	1ε
0	2ε	1ε
1ε	2ε	0
1ε	0	2ε
0	1ε	2ε

so 7 ways altogether.

Problem 4.4 Almost saturated magnet

There are 10^{21} moments antiparallel to the magnetic field out of a total of 10^{22} . The number of accessible microstates is thus

$$\Omega = \frac{10^{22}!}{10^{21}! (9 \times 10^{21})!} ,$$

and the entropy is $S = k_B \ln \Omega = k_B (\ln [10^{22}!] - \ln [10^{21}!] - \ln [9 \times 10^{21}!])$.

Using Stirling's approximation ($\ln n! \simeq n \ln n - n$ for really large n), this is

$$\begin{aligned} S &= k_B(10^{22} \ln 10^{22} - 10^{21} \ln 10^{21} - 9 \times 10^{21} \ln 9 \times 10^{21}) \\ &= 1.38 \times 10^{-23} \times 4.0 \times 10^{21} \\ &= 0.055 \text{ J K}^{-1} . \end{aligned}$$